

Vector Fields

Definition of Vector Field

A **vector field** over a plane region R is a function $\mathbf{F}$ that assigns a vector $\mathbf{F}(x, y)$ to each point in R .

A **vector field** over a solid region Q in space is a function $\mathbf{F}$ that assigns a vector $\mathbf{F}(x, y, z)$ to each point in Q .

DEFINITION Gradient Field

The **gradient field** of a differentiable function $f(x, y, z)$ is the field of gradient vectors

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}.$$

Vector Fields

The *gradient* is one example of a vector field.

For example, if

$f(x, y, z) = x^2 + y^2 + z^2$ then the *gradient* of f

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

$$= 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$$

Vector field in space

is a vector field in space.

Note that the component functions for this particular vector field are $2x$, $2y$, and $2z$.

Conservative Vector Fields

Definition of Conservative Vector Field

A vector field $\mathbf{F}$ is called **conservative** when there exists a differentiable function f such that $\mathbf{F} = \nabla f$. The function f is called the **potential function** for $\mathbf{F}$.

Curl of a Vector Field

$$\text{curl } \mathbf{F}(x, y, z) = \nabla \times \mathbf{F}(x, y, z)$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix}$$

$$= \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) \mathbf{i} - \left(\frac{\partial P}{\partial x} - \frac{\partial M}{\partial z} \right) \mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k}$$

Definition of Curl of a Vector Field

The curl of $\mathbf{F}(x, y, z) = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ is

$$\begin{aligned} \text{curl } \mathbf{F}(x, y, z) &= \nabla \times \mathbf{F}(x, y, z) \\ &= \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) \mathbf{i} - \left(\frac{\partial P}{\partial x} - \frac{\partial M}{\partial z} \right) \mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k}. \end{aligned}$$

If $\text{curl } \mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is said to be **irrotational**.

Curl of a Vector Field

Example1: Find curl $\mathbf{F}$ of the vector field

$$\mathbf{F}(x, y, z) = 2xy\mathbf{i} + (x^2 + z^2)\mathbf{j} + 2yz\mathbf{k}.$$

Is $\mathbf{F}$ irrotational?

Solution:

The curl of $\mathbf{F}$ is

$$\text{curl } \mathbf{F}(x, y, z) = \nabla \times \mathbf{F}(x, y, z) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & x^2 + z^2 & 2yz \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + z^2 & 2yz \end{vmatrix} \mathbf{i} - \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ 2xy & 2yz \end{vmatrix} \mathbf{j} + \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ 2xy & x^2 + z^2 \end{vmatrix} \mathbf{k}$$

$$= (2z - 2z)\mathbf{i} - (0 - 0)\mathbf{j} + (2x - 2x)\mathbf{k} = \mathbf{0}. \quad , \mathbf{F} = \mathbf{0}, \mathbf{F} \text{ is irrotational.}$$

Divergence of a Vector Field

Definition of Divergence of a Vector Field

The **divergence** of $\mathbf{F}(x, y) = M\mathbf{i} + N\mathbf{j}$ is

$$\operatorname{div} \mathbf{F}(x, y) = \nabla \cdot \mathbf{F}(x, y) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}.$$

Plane

The **divergence** of $\mathbf{F}(x, y, z) = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ is

$$\operatorname{div} \mathbf{F}(x, y, z) = \nabla \cdot \mathbf{F}(x, y, z) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}.$$

Space

If $\operatorname{div} \mathbf{F} = 0$, then $\mathbf{F}$ is said to be **divergence free**.

Divergence of a Vector Field

Example 3: Find the divergence at $(2, 1, -1)$ for the vector field

$$\mathbf{F}(x, y, z) = x^3y^2z\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}.$$

Solution:

The divergence of $\mathbf{F}$ is

$$\operatorname{div} \mathbf{F}(x, y, z) = \frac{\partial}{\partial x}[x^3y^2z] + \frac{\partial}{\partial y}[x^2z] + \frac{\partial}{\partial z}[x^2y] = 3x^2y^2z.$$

At the point $(2, 1, -1)$, the divergence is

$$\operatorname{div} \mathbf{F}(2, 1, -1) = 3(2^2)(1^2)(-1) = -12.$$

Line Integrals of Vector Fields

To determine the work done by the force, you need consider only that part of the force that is acting in the same direction as that in which the object is moving.

This means that at each point on C , you can consider the projection $\mathbf{F} \cdot \mathbf{T}$ of the force vector $\mathbf{F}$ onto the unit tangent vector $\mathbf{T}$.

On a small subarc of length Δs_i , the increment of work is

$$\begin{aligned}\Delta W_i &= (\text{force})(\text{distance}) \\ &\approx [\mathbf{F}(x_i, y_i, z_i) \cdot \mathbf{T}(x_i, y_i, z_i)] \Delta s_i\end{aligned}$$

where (x_i, y_i, z_i) is a point in the i^{th} sub arc.

Line Integrals of Vector Fields

Consequently, the total work done is given by the integral

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) \, ds.$$

This line integral appears in other contexts and is the basis of the following definition of the **line integral of a vector field**.

Note in the definition that

$$\begin{aligned} \mathbf{F} \cdot \mathbf{T} \, ds &= \mathbf{F} \cdot \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \|\mathbf{r}'(t)\| \, dt \\ &= \mathbf{F} \cdot \mathbf{r}'(t) \, dt \\ &= \mathbf{F} \cdot d\mathbf{r}. \end{aligned}$$

Line Integrals of Vector Fields

TABLE 16.2 Six different ways to write the work integral

$\mathbf{W} = \int_{t=a}^{t=b} \mathbf{F} \cdot \mathbf{T} \, ds$	The definition
$= \int_{t=a}^{t=b} \mathbf{F} \cdot d\mathbf{r}$	Compact differential form
$= \int_a^b \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} dt$	Expanded to include dt ; emphasizes the parameter t and velocity vector $d\mathbf{r}/dt$
$= \int_a^b \left(M \frac{dg}{dt} + N \frac{dh}{dt} + P \frac{dk}{dt} \right) dt$	Emphasizes the component functions
$= \int_a^b \left(M \frac{dx}{dt} + N \frac{dy}{dt} + P \frac{dz}{dt} \right) dt$	Abbreviates the components of $\mathbf{r}$
$= \int_a^b M \, dx + N \, dy + P \, dz$	dt 's canceled; the most common form

Line Integrals of Vector Fields

Evaluating a Work Integral

To evaluate the work integral along a smooth curve $\mathbf{r}(t)$, take these steps:

1. Evaluate $\mathbf{F}$ on the curve as a function of the parameter t .
2. Find $d\mathbf{r}/dt$
3. Integrate $\mathbf{F} \cdot d\mathbf{r}/dt$ from $t = a$ to $t = b$.



Line Integrals of Vector Fields

EXAMPLE 2 Finding Work Done by a Variable Force over a Space Curve

Find the work done by $\mathbf{F} = (y - x^2)\mathbf{i} + (z - y^2)\mathbf{j} + (x - z^2)\mathbf{k}$ over the curve $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, $0 \leq t \leq 1$, from $(0, 0, 0)$ to $(1, 1, 1)$ (Figure 16.19).

Solution First we evaluate $\mathbf{F}$ on the curve:

$$\begin{aligned}\mathbf{F} &= (y - x^2)\mathbf{i} + (z - y^2)\mathbf{j} + (x - z^2)\mathbf{k} \\ &= (\underbrace{t^2 - t^2}_0)\mathbf{i} + (t^3 - t^4)\mathbf{j} + (t - t^6)\mathbf{k}\end{aligned}$$

Then we find $d\mathbf{r}/dt$,

$$\frac{d\mathbf{r}}{dt} = \frac{d}{dt}(t\mathbf{i} + \underline{t^2}\mathbf{j} + t^3\mathbf{k}) = \mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}$$

Finally, we find $\mathbf{F} \cdot d\mathbf{r}/dt$ and integrate from $t = 0$ to $t = 1$:

$$\begin{aligned}\mathbf{F} \cdot \frac{d\mathbf{r}}{dt} &= [(t^3 - t^4)\mathbf{j} + (t - t^6)\mathbf{k}] \cdot (\mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}) \\ &= (t^3 - t^4)(2t) + (t - t^6)(3t^2) = 2t^4 - 2t^5 + 3t^3 - 3t^8,\end{aligned}$$

so

$$\begin{aligned}\text{Work} &= \int_0^1 (2t^4 - 2t^5 + 3t^3 - 3t^8) dt \\ &= \left[\frac{2}{5}t^5 - \frac{2}{6}t^6 + \frac{3}{4}t^4 - \frac{3}{9}t^9 \right]_0^1 = \frac{29}{60}.\end{aligned}$$

Line Integrals of Vector Fields

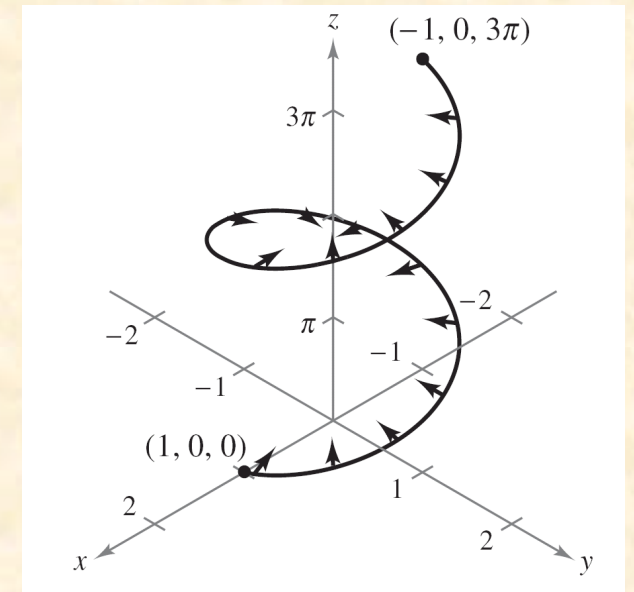
Example 4: Find the work done by the force field

$$\mathbf{F}(x, y, z) = -\frac{1}{2}x\mathbf{i} - \frac{1}{2}y\mathbf{j} + \frac{1}{4}\mathbf{k} \quad \text{Force field } \mathbf{F}$$

on a particle as it moves along the helix given by

$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k} \quad \text{Space curve } C$$

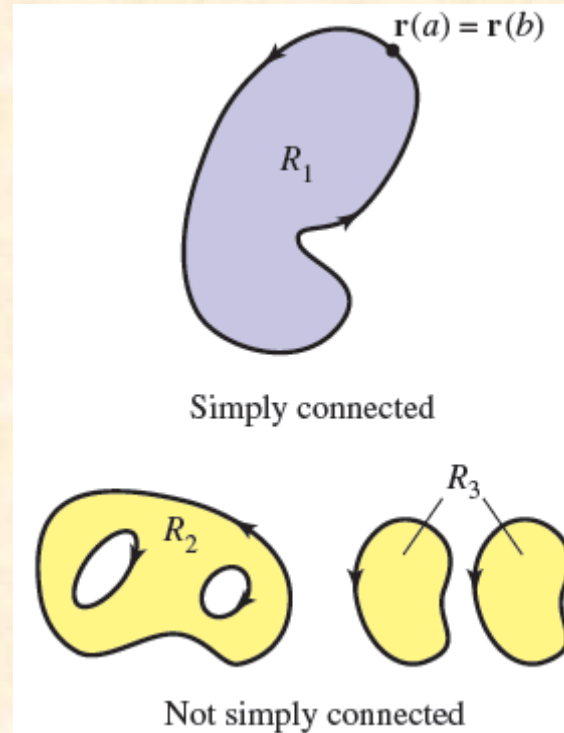
from the point $(1, 0, 0)$ to the point $(-1, 0, 3\pi)$, as shown in the Figure.



Green's Theorem

A connected plane region R is **simply connected** when every simple closed curve in R encloses only points that are in R (see Figure 15.26).

Informally, a simply connected region cannot consist of separate parts or holes.



Green's Theorem

DEFINITION Divergence (Flux Density)

The **divergence (flux density)** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is

$$\text{div } \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \quad (1)$$

If $\mathbf{F}$ denotes the velocity of a fluid in a medium. Then if $\text{div}(\mathbf{F})=0$, fluid is said to be incompressible.

In electromagnetic theory, if $\text{div}(\mathbf{F})=0$, then vector field $\mathbf{F}$ is said to be solenoidal (magnetic).

Green's Theorem

DEFINITION k-Component of Curl (Circulation Density)

The **k-component of the curl (circulation density)** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is the scalar

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}. \quad (2)$$

THEOREM 3 Green's Theorem (Flux-Divergence or Normal Form)

The outward flux of a field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ across a simple closed curve C equals the double integral of $\text{div } \mathbf{F}$ over the region R enclosed by C .

$$\oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \oint_C M \, dy - N \, dx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy \quad (3)$$

Outward flux

Divergence integral

Green's Theorem

THEOREM 4 Green's Theorem (Circulation-Curl or Tangential Form)

The counterclockwise circulation of a field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ around a simple closed curve C in the plane equals the double integral of $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$ over the region R enclosed by C .

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \oint_C M \, dx + N \, dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy \quad (4)$$

Counterclockwise circulation Curl integral

Example 5: Using Green's theorem calculates the outward flux of the vector field across the square bounded by the lines $x = \pm 1$ and $y = \pm 1$.

Green's Theorem

Solution Calculating the flux with a line integral would take four integrations, one for each side of the square. With Green's Theorem, we can change the line integral to one double integral. With $M = x$, $N = y^2$, C the square, and R the square's interior, we have

$$\begin{aligned}\text{Flux} &= \oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \oint_C M \, dy - N \, dx \\ &= \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy && \text{Green's Theorem} \\ &= \int_{-1}^1 \int_{-1}^1 (1 + 2y) \, dx \, dy = \int_{-1}^1 \left[x + 2xy \right]_{x=-1}^{x=1} dy \\ &= \int_{-1}^1 (2 + 4y) \, dy = \left[2y + 2y^2 \right]_{-1}^1 = 4. \quad \blacksquare\end{aligned}$$

Stokes' Theorem

THEOREM 5 Stokes' Theorem

The circulation of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ around the boundary C of an oriented surface S in the direction counterclockwise with respect to the surface's unit normal vector $\mathbf{n}$ equals the integral of $\nabla \times \mathbf{F} \cdot \mathbf{n}$ over S .

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma \quad (4)$$

Counterclockwise
circulation

Curl integral

Stokes' Theorem

EXAMPLE 6: A fluid of constant density rotates around the z -axis with velocity $\mathbf{F} = \omega(-y\mathbf{i} + x\mathbf{j})$, where ω is a positive constant called the *angular velocity* of the rotation (Figure 16.61). Find $\nabla \times \mathbf{F}$ and relate it to the circulation density.

Solution With $\mathbf{F} = -\omega y\mathbf{i} + \omega x\mathbf{j}$, we find the curl

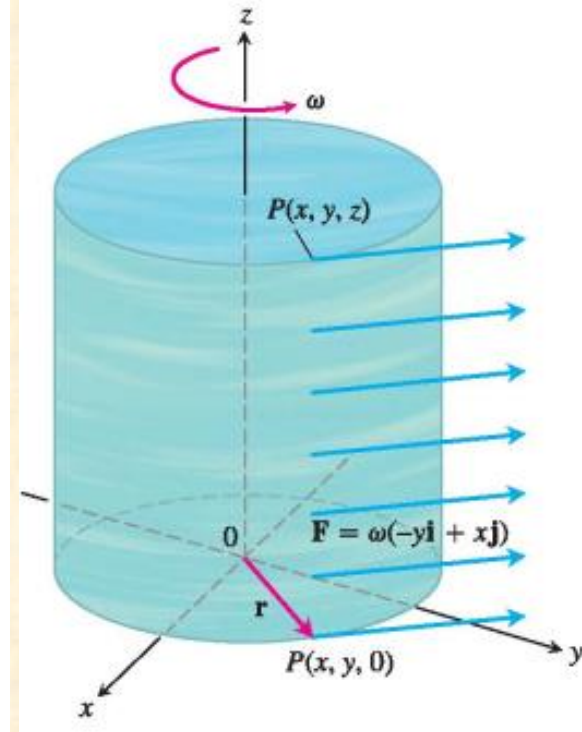
$$\begin{aligned}\nabla \times \mathbf{F} &= \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) \mathbf{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x} \right) \mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k} \\ &= (0 - 0)\mathbf{i} + (0 - 0)\mathbf{j} + (\omega - (-\omega))\mathbf{k} = 2\omega\mathbf{k}.\end{aligned}$$

By Stokes' Theorem, the circulation of $\mathbf{F}$ around a circle C of radius ρ bounding a disk S in a plane normal to $\nabla \times \mathbf{F}$, say the xy -plane, is

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iint_S 2\omega\mathbf{k} \cdot \mathbf{k} \, dx \, dy = (2\omega)(\pi\rho^2).$$

Thus solving this last equation for 2ω , we have

$$(\nabla \times \mathbf{F}) \cdot \mathbf{k} = 2\omega = \frac{1}{\pi\rho^2} \oint_C \mathbf{F} \cdot d\mathbf{r},$$



Stokes' Theorem

EXAMPLE 7: Use Stokes' Theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, if $\mathbf{F} = xzi + xyj + 3xzk$ and C is the boundary of the portion of the plane $2x + y + z = 2$ in the first octant, traversed counterclockwise as viewed from above.

Stokes' Theorem

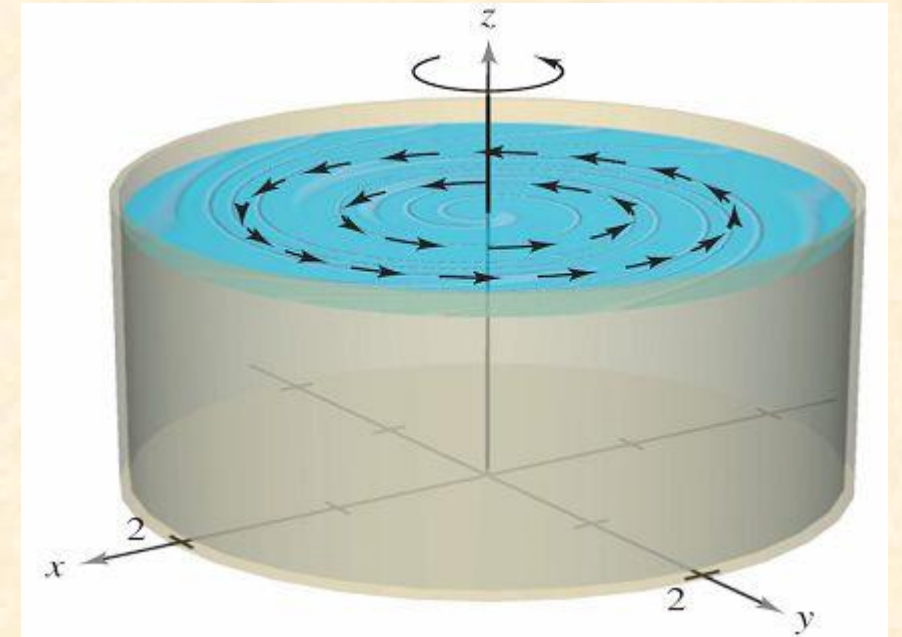
A liquid is swirling around in a cylindrical container of radius 2, so that its motion is described by the velocity field

$$\mathbf{F}(x, y, z) = -y\sqrt{x^2 + y^2}\mathbf{i} + x\sqrt{x^2 + y^2}\mathbf{j}$$

as shown in figure at right.

Find

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{N} \, dS$$



where S is the upper surface of the cylindrical container.

Stokes' Theorem

The curl of $\mathbf{F}$ is given by

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y\sqrt{x^2 + y^2} & x\sqrt{x^2 + y^2} & 0 \end{vmatrix}$$

$$= 3\sqrt{x^2 + y^2}\mathbf{k}.$$

Letting $\mathbf{N} = \mathbf{k}$, you have

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{N} \, dS = \iint_R 3\sqrt{x^2 + y^2} \, dA$$

Stokes' Theorem

$$= \int_0^{2\pi} \int_0^2 (3r)r \, dr \, d\theta$$

$$= \int_0^{2\pi} r^3 \Big|_0^2 d\theta$$

$$= \int_0^{2\pi} 8 \, d\theta$$

$$= 16\pi.$$

SUMMARY

SUMMARY OF INTEGRATION FORMULAS

Fundamental Theorem of Calculus

$$\int_a^b F'(x) dx = F(b) - F(a)$$

Green's Theorem

$$\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} dA$$

$$\int_C \mathbf{F} \cdot \mathbf{N} ds = \iiint_R \text{div } \mathbf{F} dV$$

Divergence Theorem

$$\iint_S \mathbf{F} \cdot \mathbf{N} dS = \iiint_Q \text{div } \mathbf{F} dV$$

Fundamental Theorem of Line Integrals

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(x(b), y(b)) - f(x(a), y(a))$$

Stokes's Theorem

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{N} dS$$

Thanks a lot ...